

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2096

Portland Station 138 kV
April 17, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2096. The Interconnection Customer proposes a 27.2 MW battery energy storage facility interconnecting at the Portland Station 138 kV substation in the Pennsylvania area. The request was received on May 26, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 734 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$299,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
YARDS CRK 138-1 kV Transformer Bank	CLEAR SPRINGS 500 kV/345 kV Auto	118.3	396	5.7
BULLMOOSE – MOROVIS 8801 138kV	WALNUT – WEST GALLATIN 345kV	111.8	648	5.2
NEWCOMERSTOWN – MOROVIS 8801 138kV	WALNUT – W GALLATIN SUBSTATION 345kV L	111.3	667	5.2
NEWCOMERSTOWN – VALLEY HOME TAP 2 138kV	W GALLATIN – WANUT 345kV Line	110.0	720	5.2
E ARLINGTON – VLY HOME TAP 2 138kV	WALNT – W GALLATIN 345kV	109.5	741	5.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
YARDS CRK 138-1 kV Transformer Bank	Replace transformer with higher-rated unit	\$4,444,000	4% (\$181,000)
BULLMOOSE – MOROVIS 8801 138kV	Rebuild line with higher-capacity conductor	\$2,708,000	1% (\$36,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NEWCOMERSTOWN – MOROVIS 8801 138kV	Rebuild line with higher-capacity conductor	\$3,764,000	1% (\$51,000)
NEWCOMERSTOWN – VALLEY HOME TAP 2 138kV	Reconductor line	\$1,657,000	1% (\$22,000)
EAST ARLINGTON SUBSTATION – VLY HOMETAP 2 138kV line	Reconductor line	\$696,000	1% (\$9,000)
	Total allocated to project		\$299,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.